ASSIGNMENT -1
MODULE 1

COMPUTER NETWORKS
(PCCST501)

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How the Internet Delivers a Web Page

Chosen Website: YouTube (https://www.youtube.com/)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer.

The TCP/IP protocol suite is a five-layer architecture that enables communication between devices over the Internet. It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.

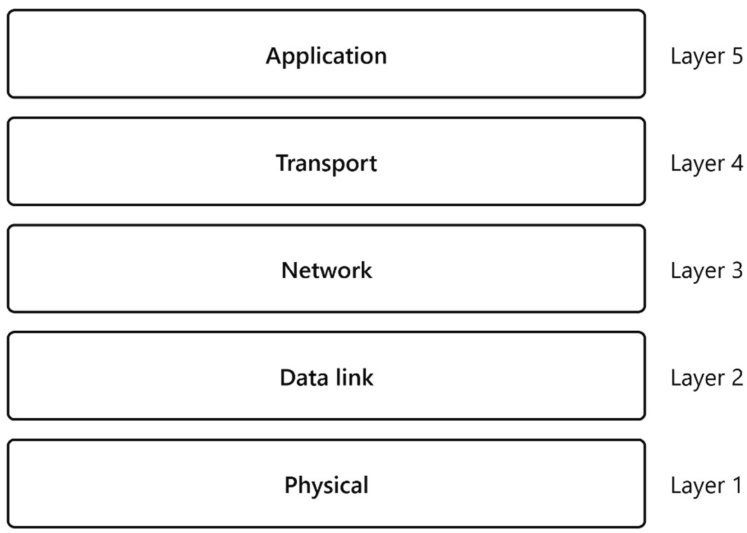

Fig 1.1: TCP/IP Protocol Stack

Application Layer

The Application layer is the topmost layer of the TCP/IP protocol suite. It provides communication between application programs running on different hosts, also known as process-to-process communication. A process sends a request to another process and receives a response. This layer contains protocols such as HTTP, HTTPS, DNS, FTP, SMTP, and DNS.

Transport Layer

The Transport layer provides end-to-end communication between the source host and the destination host. It receives messages from the Application layer, encapsulates them into transport-layer packets, and delivers them to the corresponding application on the destination host.

Network Layer

The Network layer is responsible for host-to-host communication. It also performs routing, which means selecting the best path for packets to travel from the source computer to the destination computer. The main protocol at this layer is Internet Protocol (IP).

Data Link Layer

The Data-link layer is responsible for moving the datagram across a single link. The link may be a wired LAN, wireless LAN, wired WAN, or wireless WAN. This layer encapsulates the datagram into a frame before sending it over the network.

Physical Layer

The Physical layer is responsible for carrying individual bits across the transmission medium. The transmission medium may be a copper cable, optical fibre, or wireless signals. The bits are converted into electrical, optical, or radio signals before transmission.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address.

The Domain Name System (DNS) is an application-layer protocol that helps other protocols find the network-layer address (IP address) of a computer. Instead of remembering numerical IP addresses, users can simply type easy-to-remember domain names such as www.youtube.com.

When the user enters https://www.youtube.com/ into the browser, the browser first sends a DNS request. The DNS server searches its database and returns the corresponding IP address of the YouTube server. After obtaining the IP address, the browser can establish communication with the correct server.

Without DNS, users would have to remember and type long numerical IP addresses for every website, making Internet access much more difficult. Therefore, DNS acts like the Internet's directory service by translating domain names into IP addresses.

www.youtube.com → DNS Server → IP Address

3. Describe How HTTP Is Used to Request and Receive the Webpage.

HTTP (HyperText Transfer Protocol) defines how client-server programs communicate to retrieve webpages from the Web.

After DNS returns the IP address of YouTube, the browser establishes a TCP connection with the YouTube server. Since YouTube uses secure communication, the browser sends an HTTPS GET request to request the homepage.

The YouTube server receives the request, processes it, and sends an HTTPS response containing the webpage resources such as HTML, CSS, JavaScript, images, icons, and other multimedia files.

The browser receives these resources and displays the YouTube homepage on the user's screen.

User → YouTube URL → DNS resolves IP → TCP Connection → HTTPS GET Request →

YouTube Server → HTTPS Response → Webpage Displayed.

4. Explain How TCP Provides Reliable Communication for HTTP.

TCP provides reliability in the following ways:

- **Connection Establishment:** TCP first establishes a logical connection between the client and the server before any data is transmitted.
- **Flow Control:** TCP matches the sending data rate of the source with the receiving rate of the destination so that the receiver is not overwhelmed.
- **Error Control:** TCP checks whether the received segments contain errors. If any segment is lost or corrupted, TCP retransmits it to ensure accurate delivery.
- **Congestion Control:** TCP controls the transmission rate when the network becomes congested, reducing packet loss and improving performance.

Because of these features, the YouTube webpage is received completely and in the correct order without missing or corrupted data.

5. Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer.

YouTube follows the Client–Server Architecture.

In the Client–Server architecture, two types of computers are involved: the client and the server. The client is the user's device such as computer, laptop, smartphone, which sends requests for a service. The server is a powerful computer that stores webpages, videos, and other resources and provides these services to the client whenever requested.

When the user enters https://www.youtube.com/ in a web browser, the browser acts as the client. The browser first sends a request to the YouTube server through the internet. The request passes through the layers of the TCP/IP protocol suite. After receiving the request, the YouTube server processes it, locates the requested webpage, and send s back the required data such as images, video information etc. The browser receives this information and displays the YouTube homepage to the user.

In this architecture, the client only requests services, while the server processes the requests and provides the requested resource. Multiple users can access the same YouTube server simultaneously, and each client receives its own response independently.

6. Compare HTTP and FTP.

HTTP	FTP
HTTP stands for Hypertext Transfer Protocol.	FTP stands for File Transfer Protocol.
It is mainly used for accessing web pages on the World Wide Web.	It is mainly used for transferring files between computers.
HTTP commonly uses port 80 and HTTPS uses port 443.	FTP commonly uses ports 20 and 21.
Uses one TCP connection for communication.	Uses two separate connections for communication (Control and Data Connection).
Transfers web content such as HTML pages, images, CSS, and JavaScript files.	Transfers files such as documents, images, software and multimedia files.
Used by web browsers such as Chrome, Firefox and Edge	Used by FTP client software for file transfer.

Table 6.1: Comparison Between HTTP and FTP

7. Explain How Cookies Improve User Experience with One Real-World Example.

Cookies are small text files stored by a web browser on the user's device. They help websites remember information about the user so that the browsing experience becomes faster and more personalized.

When a user logs into YouTube, cookies store information such as the login session, preferred language, dark mode setting, subscribed channels, watch history, and user preferences.

For example, suppose a user regularly watches programming videos and educational content on YouTube. The next time the user opens YouTube, cookies help the website recognize the user and display similar video recommendations without requiring the user to search again.

Similarly, if the user has already logged into YouTube, cookies help keep the user signed in, avoiding the need to enter the username and password every time the website is opened.

Thus, cookies improve convenience, save time, and provide a personalized browsing experience.

8. Flow Diagram Showing the Complete Communication.

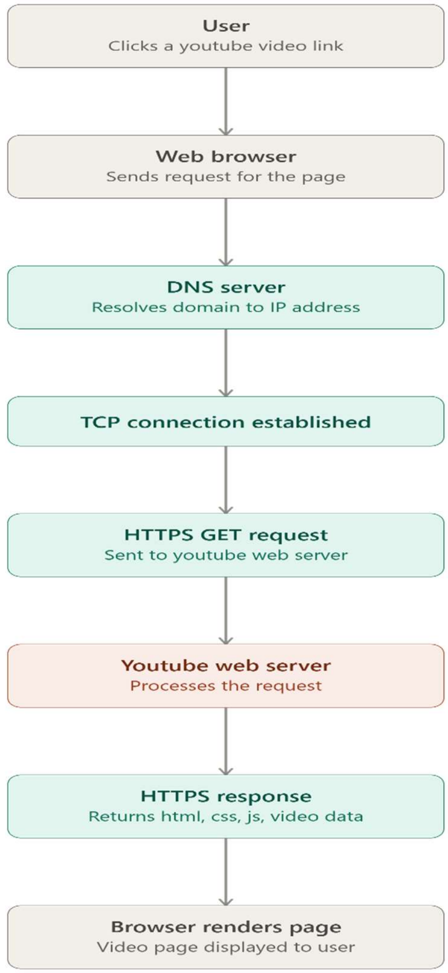

Fig 8.1: Flow Diagram of Accessing YouTube Webpage

9. Write a brief conclusion describing the importance of application-layer protocols in everyday Internet usage.

The TCP/IP protocol suite provides an organized and reliable method for communication over the Internet. When a user accesses YouTube, the Application layer uses protocols such as HTTPS and DNS to request the webpage and locate the server. The Transport layer uses TCP to establish a reliable connection and ensure that data is delivered correctly. The Network layer uses IP to route packets from the source host to the destination host.

The Data-link layer moves the datagram across each network link, while the Physical layer carries the bits through the transmission medium. All these layers work together to provide secure, reliable, and efficient communication. Therefore, application-layer protocols and the TCP/IP protocol suite play an important role in enabling users to access Internet services such as YouTube smoothly and securely.